

FEA HW5
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Question 1

Pseudocode:

```
int = integrateGeom(u,N,E)
%Calculate individual contribution of each element's integral to the total
int = 0
for i rows in E:
    %retrieve value of function at element's nodes
    for j columns in E:
        nodeVals(j) = u(E(i,:))

    %perform gauss quadrature in reference configuration (1 point exact)
    %reduces to area of element * average of nodal values (area = 0.5)
    localInt = 1/6*(sum(nodeVals))

    %Retrieve gradient of isoparametric mapping with respect to reference
    %configuration for linear triangular elements
    Bhat = [1 0 -1; 0 1 -1] %Or alternate function

    %Compute determinant of jacobian of current element
    J = Bhat*N(E(i,:),:)
    detJ = det(J)

    %Add element's contribution to total integral value
    int = int+J*localInt
```

The code can be made more or less robust to changes in basis functions or element type by creating helper functions to generate the isoparametric mapping gradient based on a specified element type. In the future, if the gradient is not a constant matrix, which occurs in many other element types, the function will also have to allow for this by increasing the degree of the Gauss quadrature to account for the degree of the determinant of the Jacobian as well.

Question 2

Code attached.

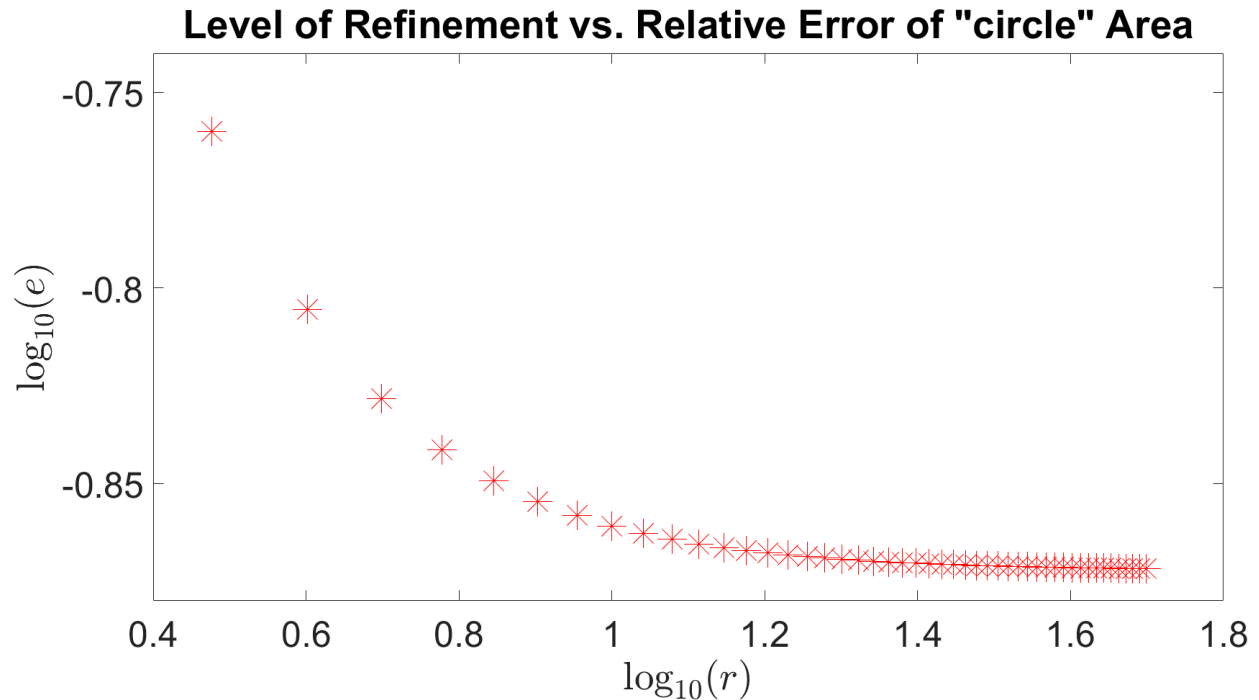
Question 3

Code attached.

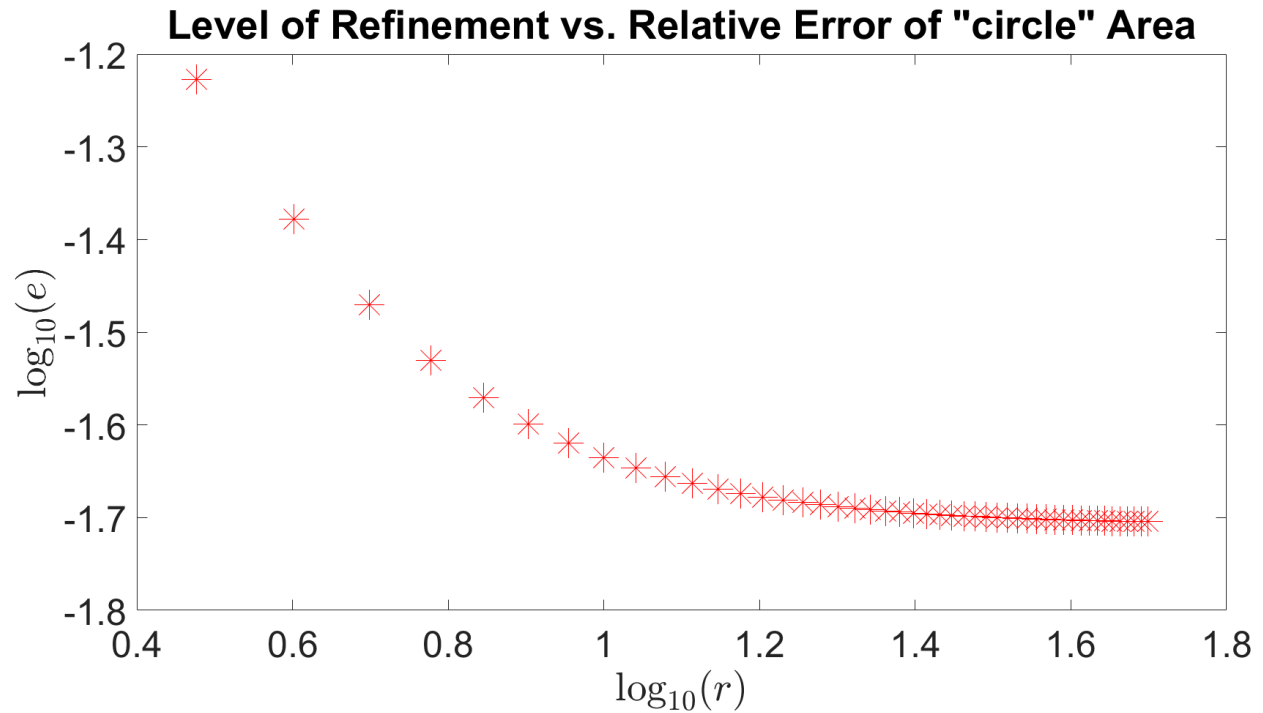
Question 4

(a)

Error was calculated as a percentage of the theoretical value, a circle of radius 10. The loglog plot was generated with mesh refinements levels from 3 to 50. First we will plot the results as is:



The first thing to notice is that error only seems to converge to a number above 10% (in fact the lowest error was 13.4%). This seems quite high, and upon inspection it was noticed that a clockwise-wound element in the original given geometry was generating incorrect values. The two choices here are either 1. to change the node ordering of this bad element or 2. to account for negative jacobian determinants in `integrateGeom.m`. Both approaches have their own merit; it was chosen to flip the incorrectly wound element in the original geometry and once again generate results:



Error now drops to under 2% (1.97% at 50 mesh refinements). This is exactly the range of error that is expected when calculating the error based on true circle area, since as mentioned in the past week, the shape is not a true circle. The value of the integral seems to converge to around 307.9552

(b)

One Gauss integration point was used since elements were linear and therefore this results in a perfect approximation. Using additional Gauss points would also be exact and therefore would not provide better convergence.